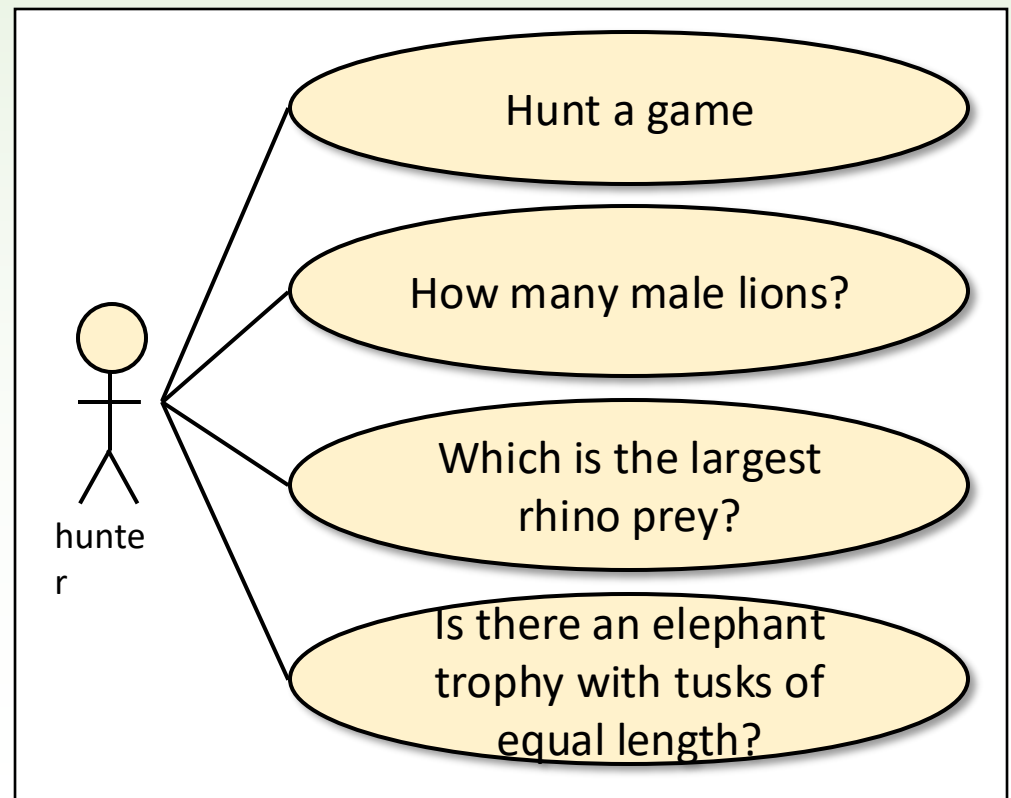


Hunting

A hunter (name, age) has collected several trophies. A trophy displays some data of the hunted game: the species (elephant, rhinoceros, lion), the location and date of the hunt, the weight of the hunted game (in kg), and some further special data (in the case of elephants, the length of each tusk separately (in cm); for rhinoceroses, the weight of the horn; for lions, the gender of the game).

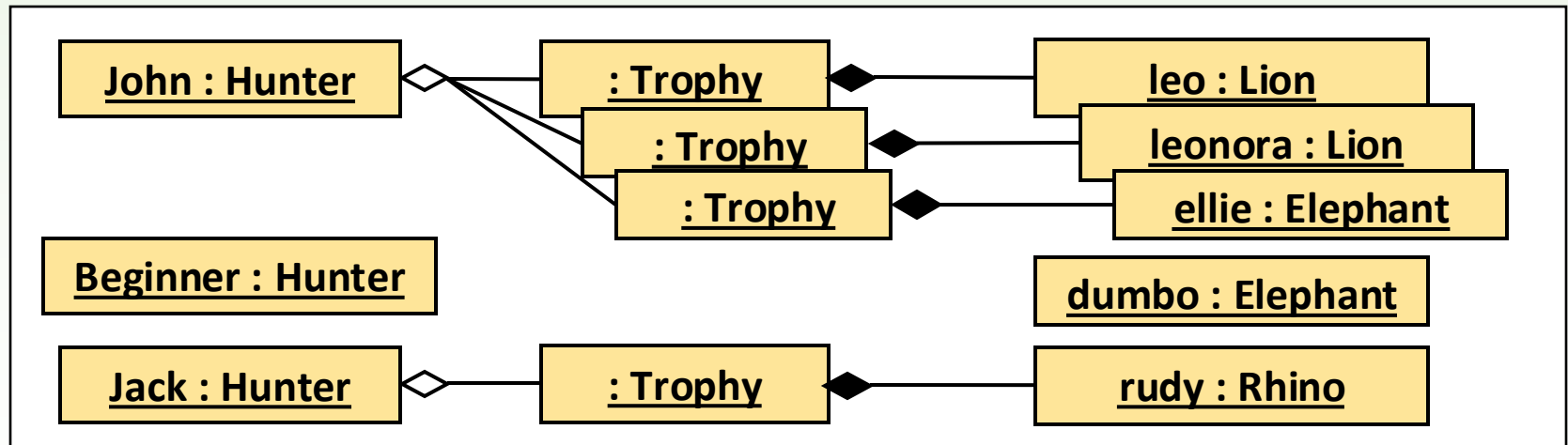
Let's answer the following questions:

- How many male lions did a hunter shoot?
- Which rhinoceros trophy of a hunter has the highest horn-to-body weight ratio?
- Does a hunter have an elephant trophy with tusks of equal length?



Hunting

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Hunting

```
Game(w:real, g:Gender)
{
  weight:=w;
  gender := g;
}
```

```
<<enumeration>>
Gender
male
female
```

Notation:

The “...” notation in a constructor's parameter list indicates that initializing parameters are provided for all attributes, so that they can be assigned as values to the corresponding data members.

Notation:

From now on, methods in the form of “IsAlosztály()” will be used in our models without the need to explicitly define them.

```
<<abstract>>
Game
# weight : real
# gender : Gender
# Game( ... )
+ isElephant():bool {virtual}
+ isRhino():bool {virtual}
+ isLion() : bool {virtual}
```

return false

return false

return false

Elephant

- left : real
- right : real

Game(w, g)
left, right := l, r

+ Elephant(w,l,r:real, g:Gender)
+ isElephant():bool {override}

return true

Rhino

horn : real

Game(w, g)
horn:= h

+ Rhino(w,h:real, g:Gender)
+ isRhino():bool {override}

return true

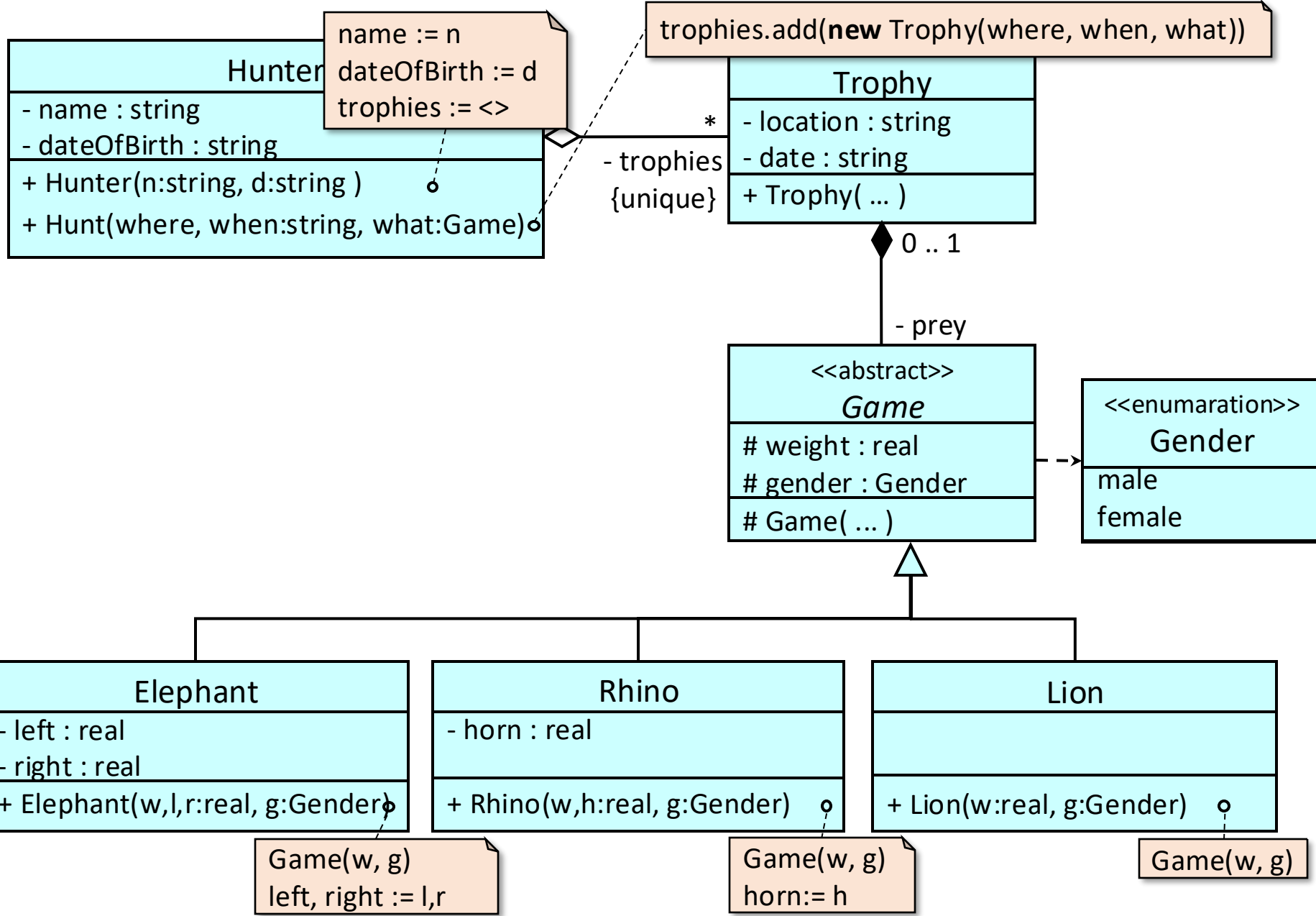
Lion

Game(w, g)

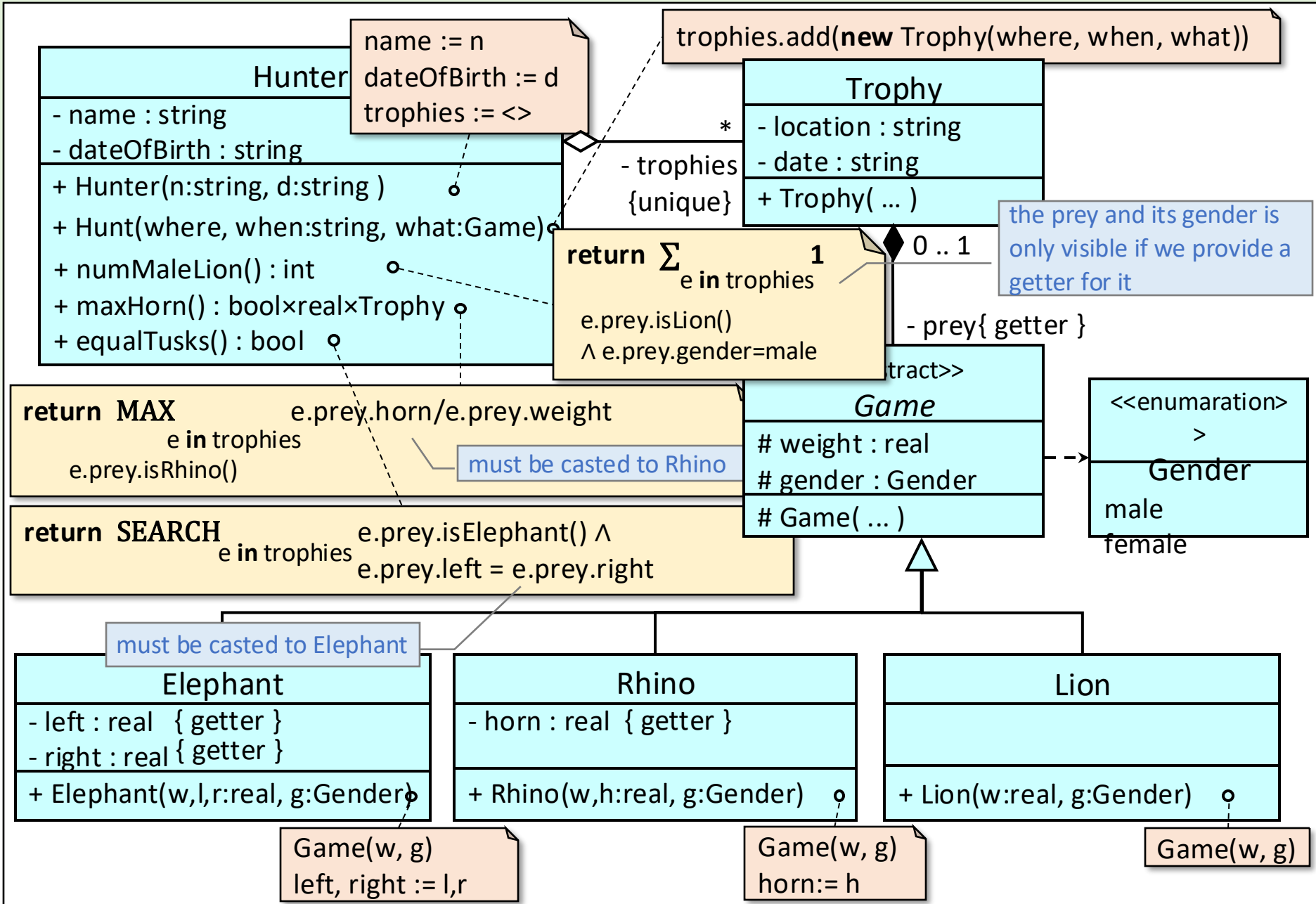
+ Lion(t:real, n:Nem)
+ isLion():bool {override}

return true

Hunting



Hunting



The fallen legions are on their way to the central star system of the Empire under the leadership of Horus Lupercal, to overthrow the Emperor's rule.

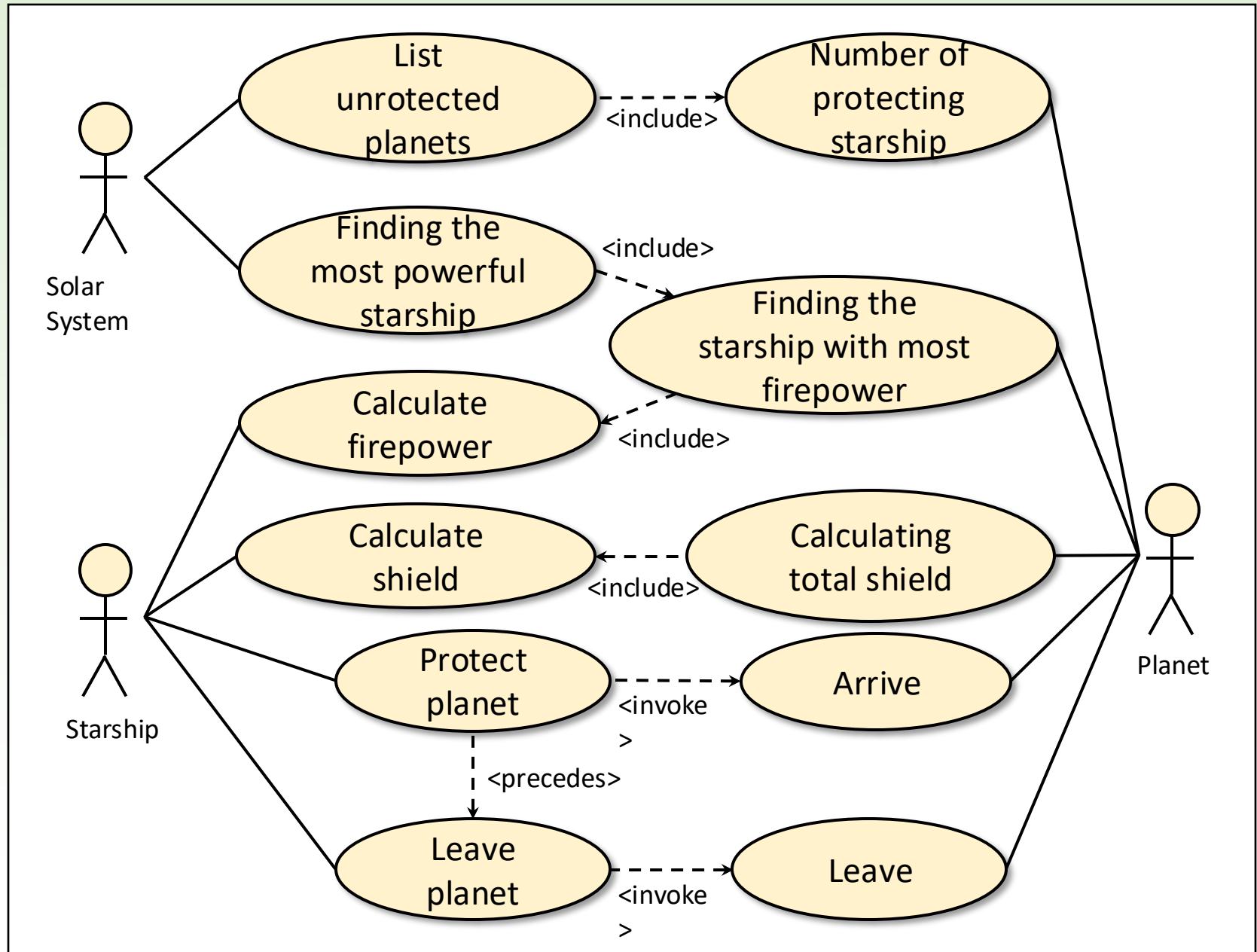
As the traitorous army approaches, the planets loyal to the Empire are plunged into darkness one by one across the star system.

Star War

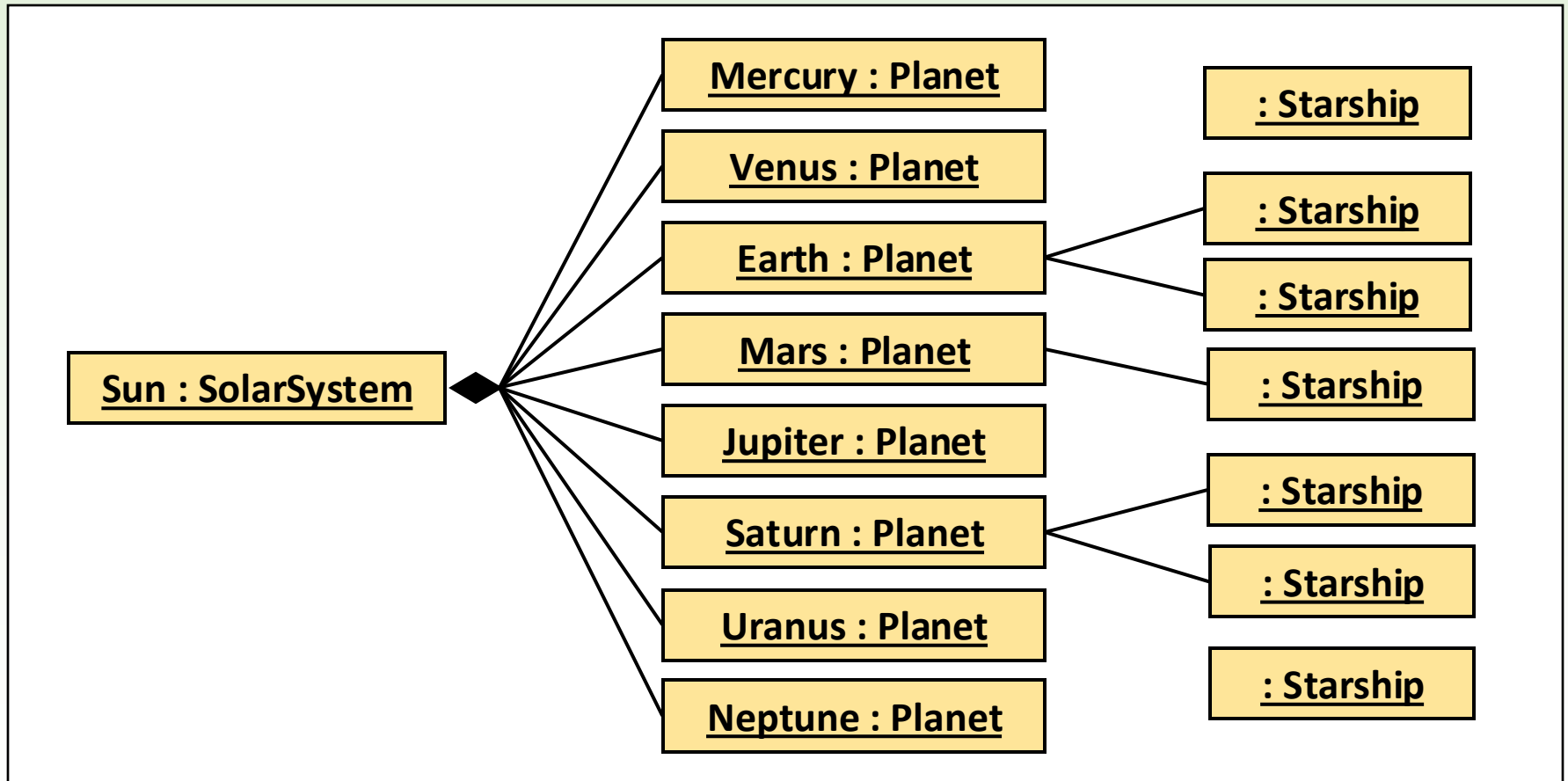
The planets in the solar system are defended by starships. There are three types of starship: star destroyers, landing ships, and laser ships. A starship's combat capability is determined by its armor (integer), its shield strength (integer), and the number of space marines serving on board:

- Star destroyer: Its firepower is equal to half of its armor.
 - Landing ship: Its firepower is equal to the number of space marines on board.
 - Laser ship: Its firepower is equal to its shield strength.
1. Which planets are not defended by any starship?
 2. Find the starship with the greatest firepower in the solar system!
 3. What is the total shield strength of the starships stationed on a given planet?

Star War



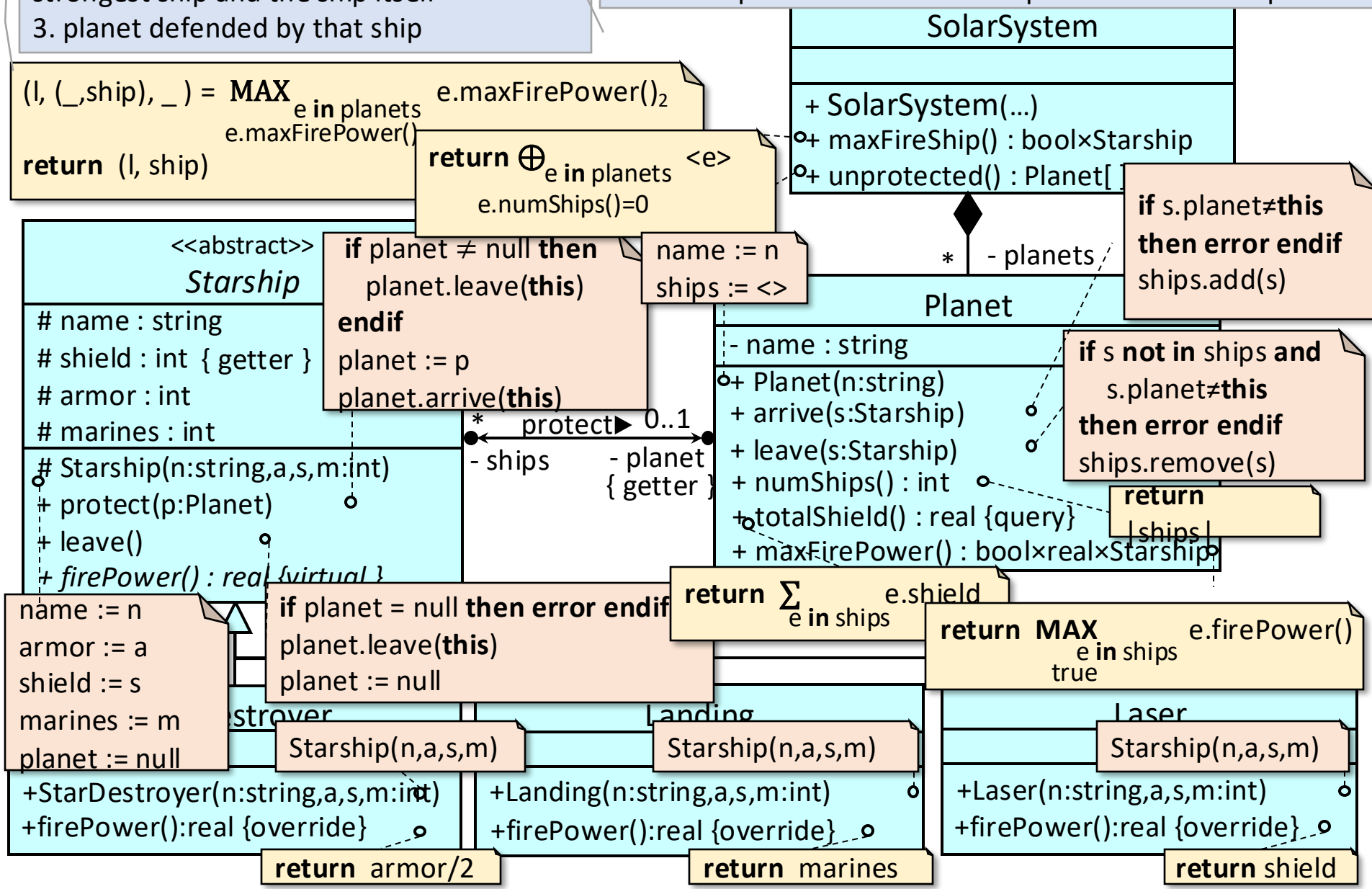
Star War



MAX return values:

1. I tells if there is a ship in the solar system
2. pair of max values: firepower of strongest ship and the ship itself
3. planet defended by that ship

MaxFirepower() returns 3 values: 1. is there a ship at the planet; 2. firepower of the strongest ship; 3. the ship itself. The 1st will be the condition of cond. max search. The other two form pairs that must be compared based on firepower.



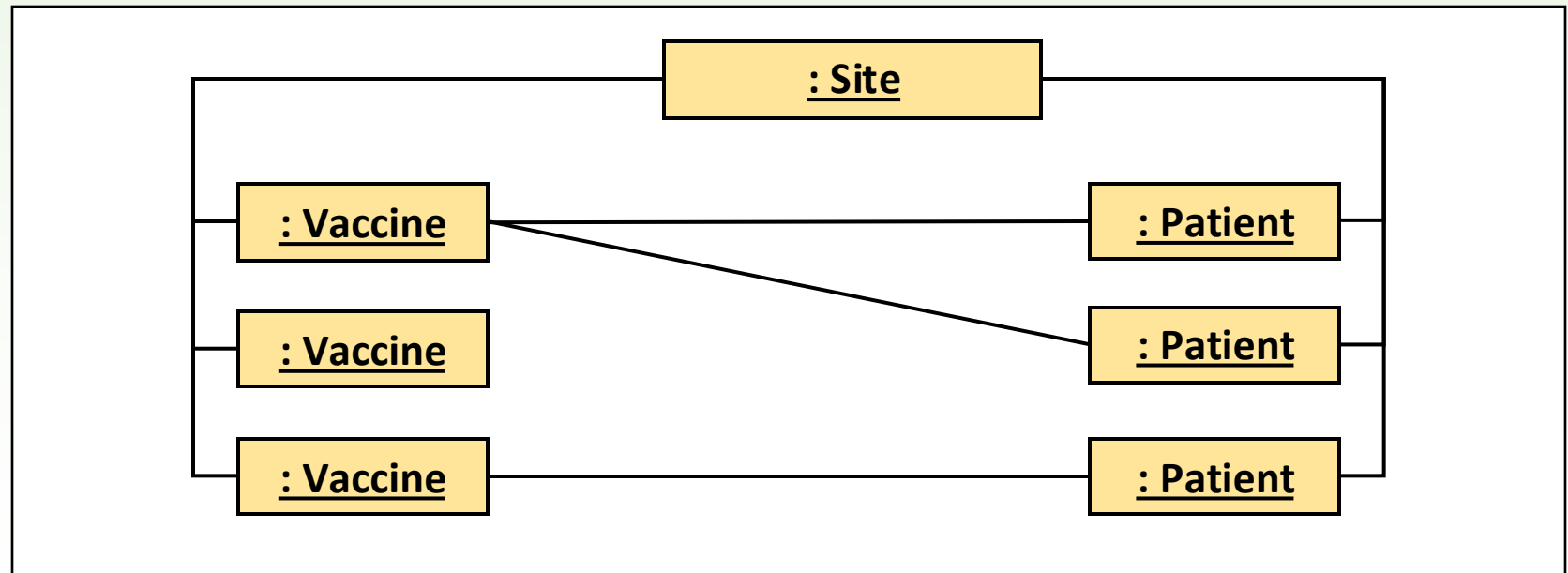
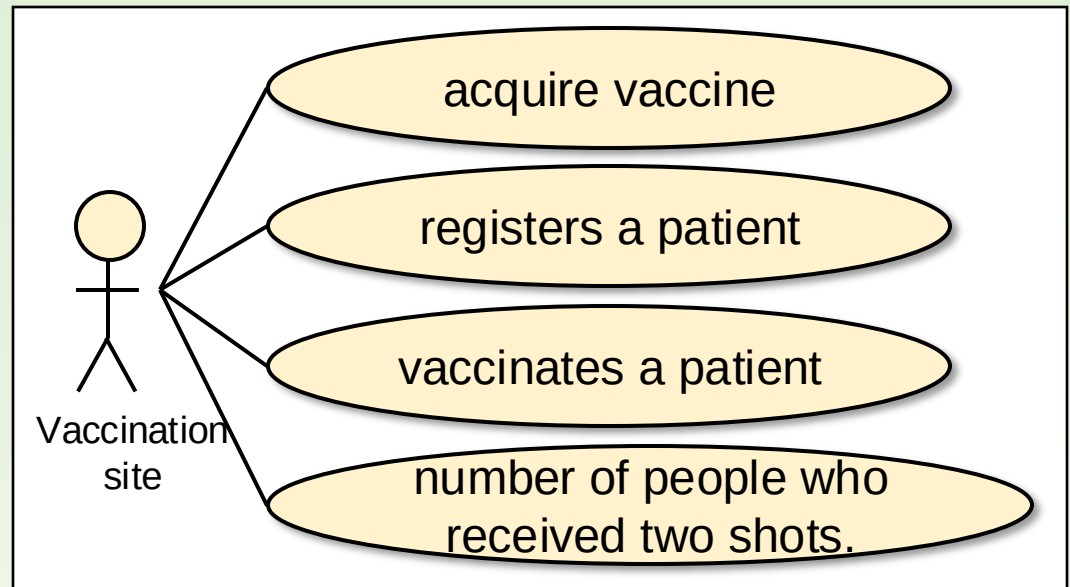
Vaccination

If a citizen wants to get vaccinated, they must first register at a vaccination site, then receive the vaccine at a specified time, provided that the vaccine chosen by the patient is available. The patient can get vaccinated multiple times with different types of vaccines. The location of the vaccination site is known, as well as how many usable doses of each type of vaccine are available. The effectiveness of each vaccine is calculated based on the number of days elapsed since the previous vaccination. For MediProduct, it's twice the days elapsed, for MediPlus, its ten plus the days elapsed, and for MediPower, its 1.2 to the power of the days elapsed. For each vaccine, the maximum effectiveness is 50. We keep records of when vaccinated patients received which vaccines.

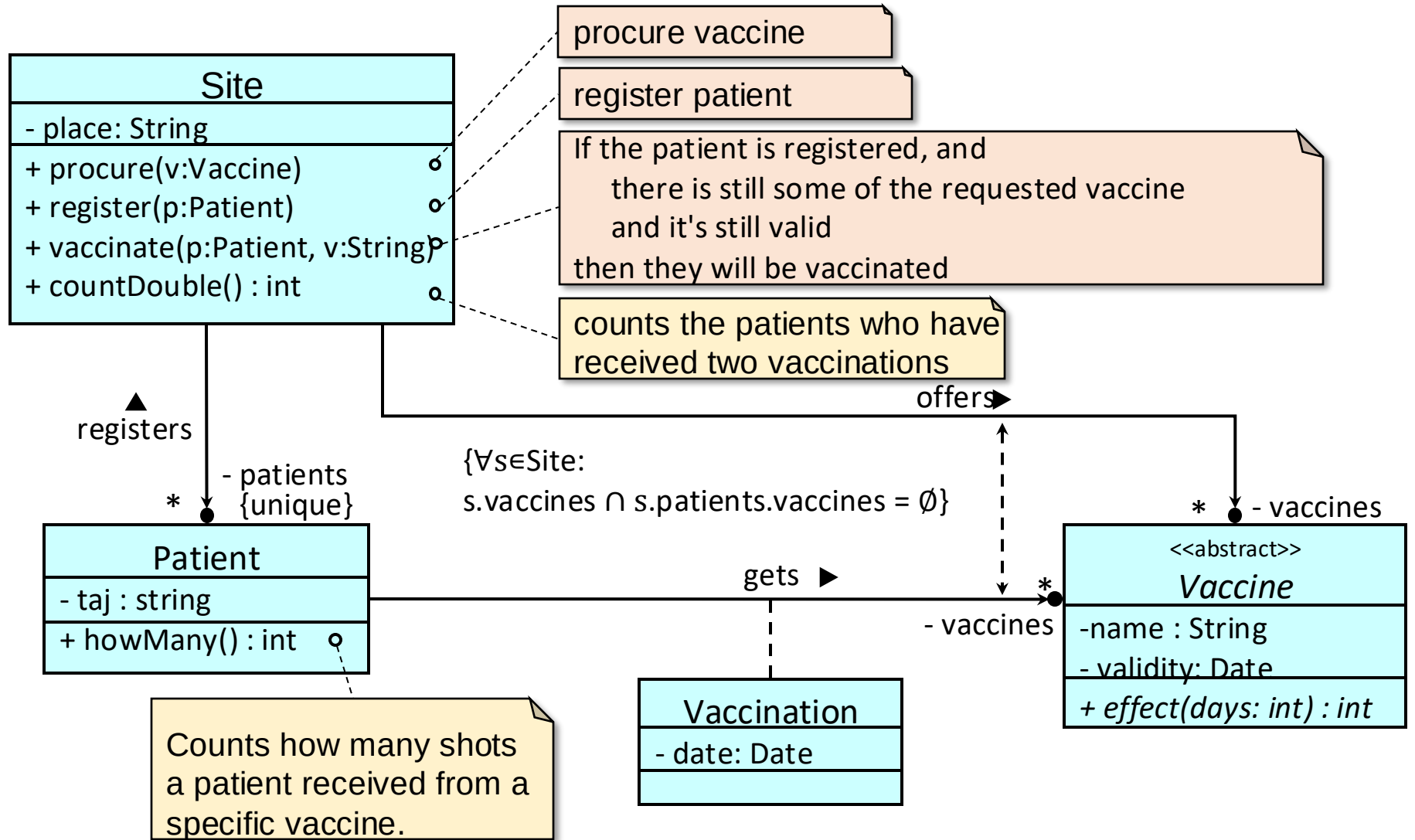
Features:

- a patient can register at the vaccination site
- the vaccination site should be able to receive vaccines.
- it should be possible to record when a registered patient gets vaccinated with a vaccine of their choosing.
- be able to query how many patients have received their second vaccination.

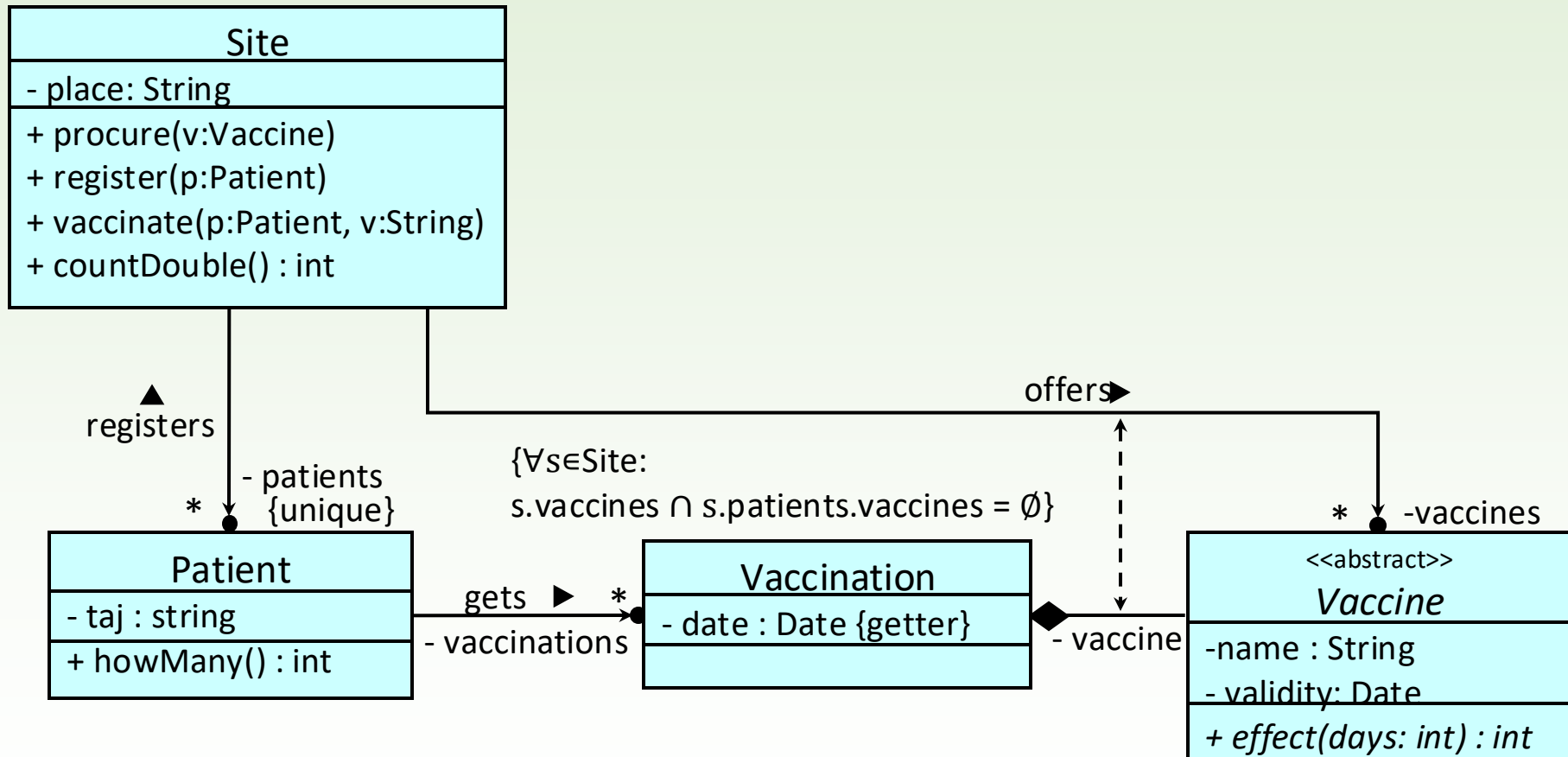
Vaccination



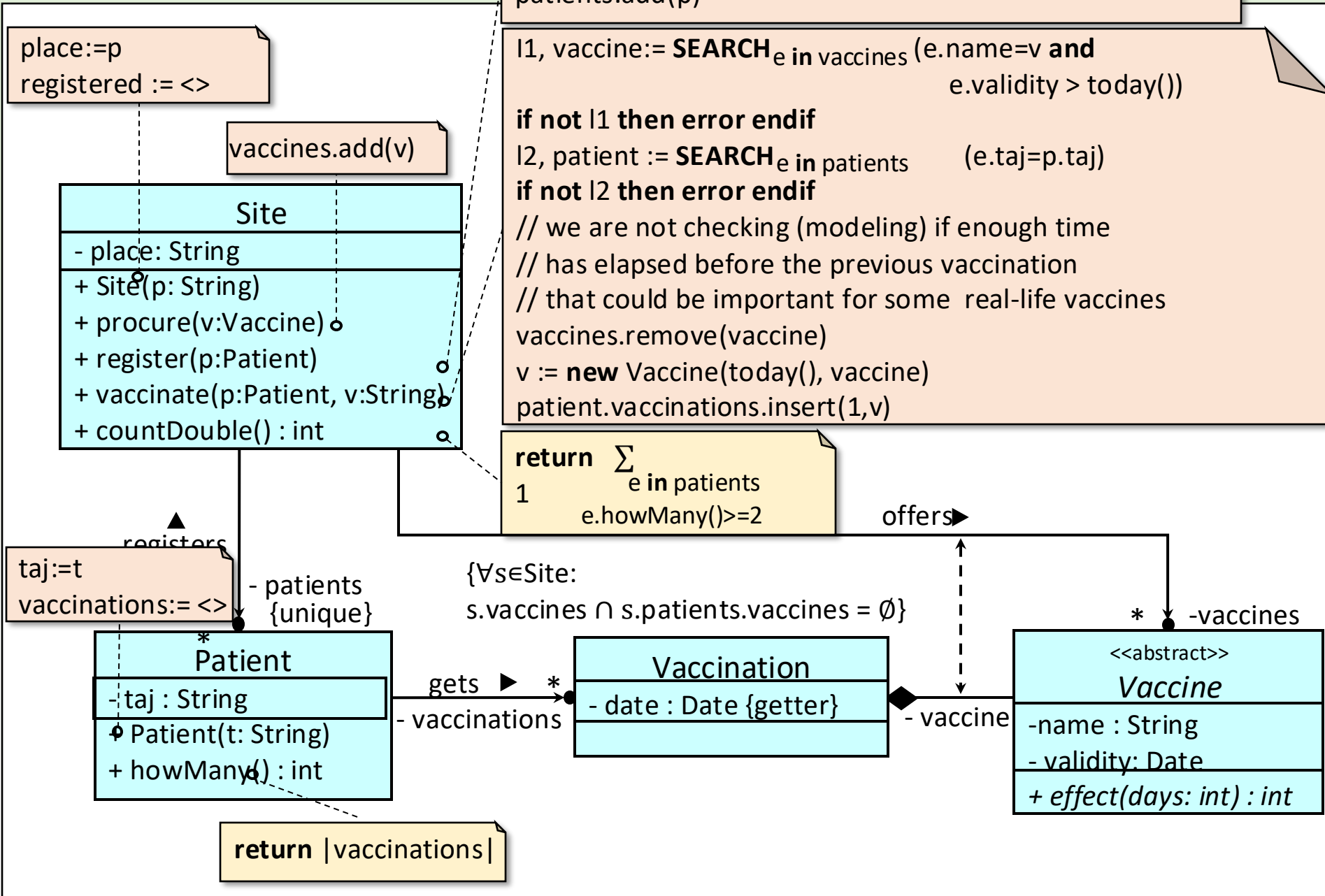
Vaccination



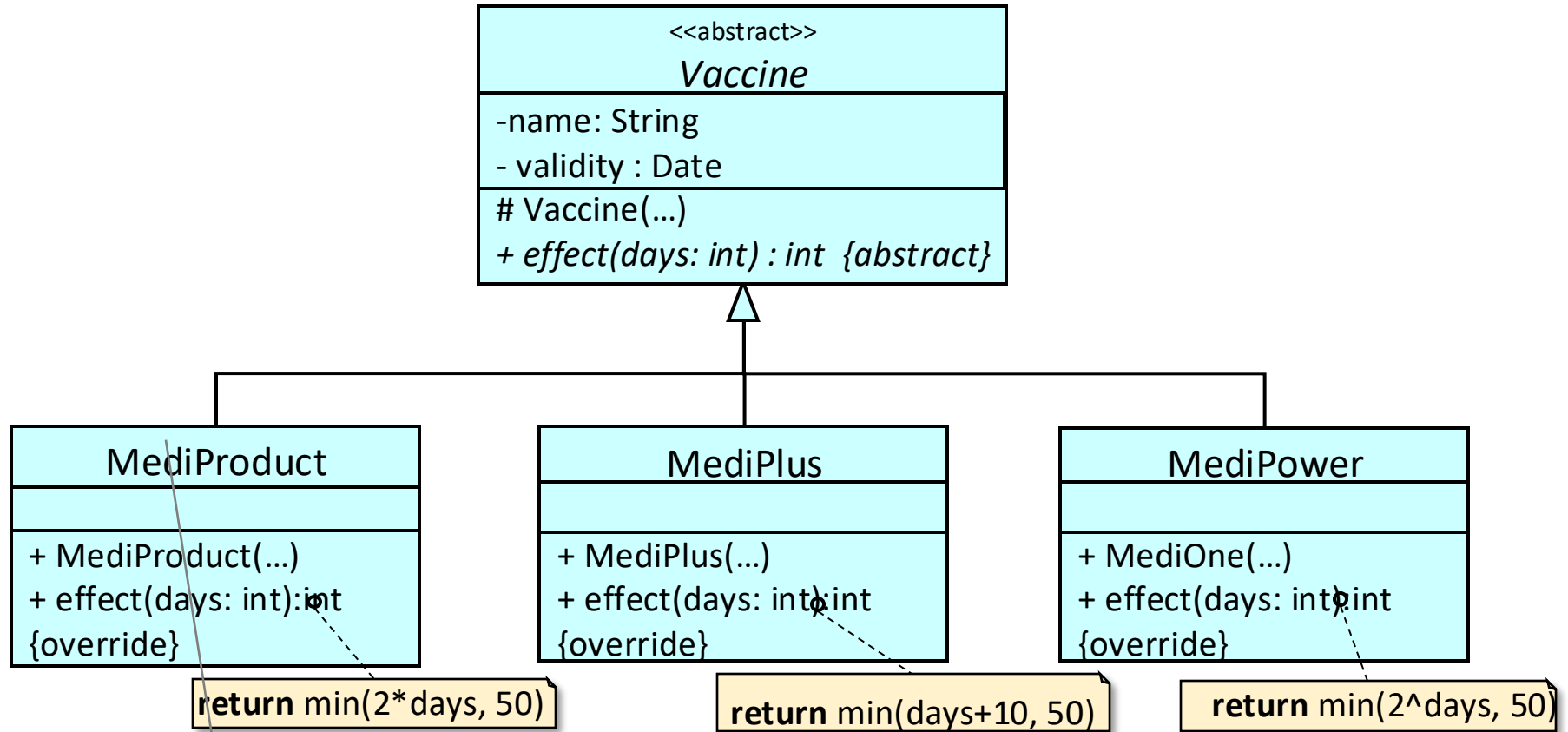
Vaccination



Vaccination



Vaccination



The constructor just calls the parent class's constructor, passing along its parameters (in this case, just the expiration date), and sets the name of the vaccine.